Aerospace Engineering

	Normative 4-year Degree
Major	Aerospace Engineering 航空航天工程
Degree / Award Title	Bachelor of Engineering in Aerospace Engineering 工學士(航空航天工程)
Offering Academic Unit	Department of Mechanical Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12
College-specified Courses	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3

	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	MNE2066 Engineers in Society	3

College / School Requirements (6 credit units) **Catalogue Term : Semester A 2025/26**

Course Code	Course Title	Credit Units
CS1302	Introduction to Computer Programming	3
PHY1101	Introductory Classical Mechanics	3

Major Requirements (84 credit units) **Catalogue Term : Semester A 2026/27**

1. Major Core Courses (81 credit units)

a) Core Courses (69 credit units)

Course Code	Course Title	Credit Units	Remarks
MA2001	Multi-variable Calculus and Linear Algebra	3	
MA3001	Differential Equations	3	
MNE2020	Engineering Workshop Practice	0	
MNE2029	Electrical and Electronic Principles I	3	
MNE2036	Engineering Computing	3	
MNE2109	Engineering Mechanics	3	
MNE2110	Engineering Materials	3	
MNE2112	Thermodynamics	3	
MNE2116	Engineering Graphics	0	# Department-specified Course for top 40% majors allocation
MNE2204	Aircraft System Design	3	
MNE3049	Control Principles	3	
MNE3118	Mechanics of Materials	3	
MNE3122	Fluid Mechanics	3	
MNE3202	Aircraft Dynamics	3	
MNE3203	Aerospace Propulsion Systems	3	
MNE3204	NDT Technologies for Aircraft Structures and Materials	3	
MNE3205	Flight Mechanics	3	
MNE4109	Reliability Engineering and Risk Analysis	3	
MNE4201	Aerodynamics	3	
MNE4202	Aeroelasticity	3	

MNE4203	Modelling and Simulation	3	
PHY1202	General Physics II	3	# Department-specified Course for top 40% majors allocation

Select **ONE** from the following:

Course Code	Course Title	Credit Units	Remarks
MNE4068	Project (Individual)	9	A course to fulfil the internship/ consultancy project/research project requirement
MNE4119	Project (Group)	9	A course to fulfil the internship/ consultancy project/research project requirement

b) Aerospace Materials/ Aerospace Systems Stream Courses (12 credit units)

Aerospace Materials Stream (12 credit units)
Select four courses from the following:

Course Code	Course Title	Credit Units
MNE3119	Manufacturing Technology	3
MNE3121	Heat Transfer	3
MNE3123	Internship in Engineering	3
MNE4010	Dynamics and Vibrations	3
MNE4121	Machine Learning and Quantum Computation	3
MNE4208	Advanced Composites and Polymers	3
MNE4213	Advanced Structural Materials	3

OR

Aerospace Systems Stream (12 credit units)
Select four courses from the following:

Course Code	Course Title	Credit Units
MNE3121	Heat Transfer	3
MNE3123	Internship in Engineering	3
MNE3207	Avionic Power Systems	3
MNE4010	Dynamics and Vibrations	3
MNE4121	Machine Learning and Quantum Computation	3
MNE4204	Signals, Sensors and Communication Systems	3
MNE4217	Autonomous Systems and Navigation	3

OR

For students who do not want to focus on a selected stream, they can take any **four** courses from
the following list (12 credit units).

Course Code	Course Title	Credit Units
MNE3119	Manufacturing Technology	3
MNE3121	Heat Transfer	3
MNE3123	Internship in Engineering	3
MNE3207	Avionic Power Systems	3
MNE4010	Dynamics and Vibrations	3
MNE4121	Machine Learning and Quantum Computation	3
MNE4204	Signals, Sensors and Communication Systems	3
MNE4208	Advanced Composites and Polymers	3
MNE4213	Advanced Structural Materials	3
MNE4217	Autonomous Systems and Navigation	3

2. Electives (3 credit units)

Course Code	Course Title	Credit Units
CS2311	Computer Programming	3
FS2001	Workshop-based Study in Science and Engineering	3
MNE2114	Experimental Sustainable Engineering Techniques	3
MNE3046	Automation Technology	3
MNE3104	Sustainable and Renewable Energy	3
MNE3125	Fluid Mechanics for Energy-Related Applications	3
MNE3126	Methodologies for Solving Complex Engineering Problems	3
MNE3127	Electron Microscopy	3
MNE3128	Radiological Analysis	3
MNE4005	Finite Element Analysis	3
MNE4032	Robotics and Machine Vision	3
MNE4047	Directed Studies	3
MNE4125	Principles and Transport Process Fundamentals of Semiconductor Manufacturing	3
SYE3102	Quality Engineering	3
SYE4024	Project Management	3

3. Optional Courses

Course Code	Course Title	Credit Units	Remarks
FS2002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS3002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS4001	Co-operative Education Scheme (CES)	8	Internship (8 - 12 months)
FS4002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS4005	Overseas Internship Scheme	3	Internship (9 - 13 weeks)